Notes: Myers (JFE, 1977)

Determinants of Corporate Borrowing

Contents

1 Big picture

- **Question.** Why do firms not borrow "as much as possible," even when debt may carry tax advantages? What determines debt capacity, debt maturity, and the sensible upper limit to borrowing?
- **Core contribution.** Myers shows that debt can destroy firm value by distorting future investment decisions. When a firm has valuable future investment opportunities, existing risky debt transfers part of the gain from those investments to bondholders. Equity holders then underinvest.
- **Main idea.** Many corporate assets are really *real options*: their value depends on future discretionary expenditure by the firm. Growth opportunities, maintenance, marketing, R&D, training, and even keeping a going concern alive all require future investment choices. Risky debt changes those choices.
- **Baseline mechanism.** Suppose shareholders must invest I in the future to capture an asset worth $V(s)$ in state s. With all-equity financing they invest whenever $V(s) \geq I$. With risky long-term debt outstanding, shareholders invest only when $V(s) \geq I + P$, where P is the promised payment to creditors. Positive-NPV states between I and $I + P$ are abandoned.
- **Immediate implication.** Debt reduces the present market value of firms with growth opportunities because it induces *suboptimal underinvestment*. In modern language, this is the classic debt-overhang logic.
- **What the paper does *not* need.** The argument does not rely on bankruptcy costs, incomplete financial markets, or managerial empire-building. Myers deliberately shows that the distortion survives even in a world with perfect and complete financial markets.
- **Main prediction for leverage.** Optimal debt is lower when a larger fraction of firm value comes from future growth opportunities or other payoffs contingent on discretionary future spending. Assets already in place support more debt than options on future assets.
- **Why book-value leverage can make sense.** Practitioners often judge debt capacity against book value or tangible capital rather than total market value. Myers rationalizes this: book value is closer to assets already in place, whereas market value includes growth opportunities that support less debt.
- **Credit-rationing result.** The amount a firm can borrow is not the same as the amount it *should* borrow. There is a maximum market value of debt available, and raising the promised payment beyond a point can actually *reduce* debt value.
- **Maturity-matching result.** The paper rationalizes matching the maturities of assets and liabilities. Long-maturity debt against short-lived or rapidly shrinking assets worsens the incentive problem because debt claims outlast the asset value currently in place.
- **Target debt-ratio result.** Debt tied to assets in place, or debt limits that move with the value of those assets, can mitigate the agency problem. This gives a rationale for target debt ratios.
- **Key nuance.** Risk itself is not the right summary statistic. A firm may be risky and still support debt if its assets are mostly in place. What matters is whether value depends on future discretionary investment and how much of the gain from that investment leaks to bondholders.

- **Diversification result.** Combining projects or firms does not automatically increase debt capacity. Depending on how project payoffs line up with debt burdens, diversification can help, do nothing, or hurt.
- **Tax appendix.** Corporate tax shields can justify positive debt, but only up to the point where the tax benefit is offset by the loss from distorted future investment. The value-maximizing firm stops before the maximum amount of debt that markets would be willing to price.
- **Takeaway.** Corporate borrowing is constrained not just by default risk but by the interaction between risky debt and future real investment. The more a firm's value is an option on future action, the less debt it should carry.

2 Data and measurement (key objects)

2.1 This is a theory paper: the "objects" are model primitives

There is no estimation sample, regression table, or identification design in the modern empirical sense. The key objects are state-contingent firm values, investment thresholds, debt promises, and the decomposition of firm value into assets in place and growth opportunities.

2.2 Firm value decomposition

The paper starts from the decomposition

$$V = V_A + V_G, \tag{1}$$

where:

- V is total firm value,
- V_A is the value of assets already in place,
- V_G is the present value of future growth opportunities.

Interpretation. V_G is not just "future profits." It is better viewed as the value of *options* to make future investments on favorable terms.

2.3 State variable and Arrow–Debreu prices

The future state of nature is denoted s. The price today of one dollar delivered next period only in state s is $q(s)$.

Interpretation. Myers works in a complete-markets setting. Using state prices lets him show that the borrowing problem is not a hidden artifact of financial-market incompleteness.

2.4 The single-option baseline

The simplest firm has:

- no assets in place initially, so $V_A = 0$,
- one future investment opportunity,
- required future outlay I,
- payoff from exercising the option equal to $V(s)$ in state s.

If the firm does not invest, the option expires and the firm is worthless.

Interpretation. This is the cleanest possible setup for isolating how existing debt affects future exercise of a positive-NPV investment option.

2.5 Investment decision rule

Let $x(s) \in \{0, 1\}$ denote whether the firm invests in state s.

- $x(s) = 1$ means the option is exercised.
- $x(s) = 0$ means the option is abandoned.

Under all-equity financing, investment occurs whenever $V(s) \geq I$.

2.6 Threshold states

Myers uses threshold states to summarize the investment rule.

- s_a is the all-equity breakeven state, defined by $V(s_a) = I$.
- s_b is the levered breakeven state when long-term debt is outstanding, defined by $V(s_b) = I + P$ in the simple no-tax case.

Interpretation. Debt shifts the cutoff from s_a to s_b. That shift is the source of underinvestment.

2.7 Debt and equity values

The market value of debt is V_D and the market value of equity is V_E, with

$$V = V_D + V_E. \tag{2}$$

The debt contract promises payment P at maturity.

2.8 Timing matters

The paper distinguishes two debt-timing cases:

- **Short-maturity debt:** debt matures after the state is revealed but *before* the firm decides whether to invest.
- **Long-maturity debt:** debt matures *after* the investment option expires.

Interpretation. Only the second case creates the underinvestment problem in the simple model, because equity holders know bondholders will share the upside if they invest.

2.9 Real assets versus real options

In the generalization, Myers distinguishes:

- **real assets:** assets whose market value does not depend on further discretionary investment,
- **real options:** opportunities to acquire or preserve real assets by future discretionary spending.

Interpretation. This is the deeper version of the assets-in-place versus growth-opportunity split. The point is not simply "old" versus "new" assets; it is whether value depends on future action.

2.10 Multiperiod objects

In the multiperiod discussion, the key variables are:

- I_t: incremental discretionary investment at time t,
- V_t: market value of the firm at time t,
- $V_{E,t}$ and $V_{D,t}$: market values of equity and debt at time t,
- $\sigma_t^2 \equiv \sigma^2(\tilde{V}_{t+1}/V_t)$: the variance rate of overall market value.

Interpretation. The debt problem is not one-period only. Once debt distorts one investment decision, it changes future firm value and future debt risk, generating spillovers across time.

2.11 Transfer to bondholders

A central conceptual object later in the paper is

$$Z_t \equiv \frac{dV_t}{dI_t} - \frac{dV_{E,t}}{dI_t}, \tag{3}$$

which measures how much of the value created by shareholders' investment leaks to existing bondholders.

Interpretation. If $Z_t > 0$, shareholders bear the full cost of investment but do not capture the full benefit. Then underinvestment is natural.

2.12 Secondary-market/liquidation value

The paper also studies real assets that can be sold in secondary markets. The liquidation value plays the same formal role as the exercise price in the growth-option example.

Interpretation. The model applies not only to future investment but also to disinvestment or abandonment decisions.

2.13 Tax objects in the appendix

The appendix introduces:

- T: the corporate tax rate,
- R: a maximum promised interest rate that may bound deductibility,
- tax shields generated by the interest expense $P - V_D$.

Interpretation. Taxes make some debt worthwhile, but they do not overturn the central real-options agency cost.

3 Models and empirical specifications (all equations + interpretation)

3.1 Equation (1): firm value as assets in place plus growth opportunities

$$V = V_A + V_G. \tag{1}$$

Interpretation.

- V_A is the portion of value already "locked in" by assets in place.
- V_G is the value of future discretionary investment opportunities.
- The paper's core claim is that V_G supports less debt than V_A because future investment can be discouraged by risky debt.

3.2 Equation (2): general state-price valuation of the option-driven firm

$$V = \int_0^\infty q(s)\, x(s)\, [V(s) - I]\, ds. \tag{2}$$

Interpretation.

- The firm contributes value only in states where it actually invests, that is, where $x(s) = 1$.
- The term $V(s) - I$ is the state-contingent net present value of exercise at the decision date.
- This equation makes the future investment rule itself part of current firm value.

3.3 Equation (3): all-equity financing

Under all-equity financing, shareholders invest whenever $V(s) \geq I$. Let s_a satisfy $V(s_a) = I$. Then

$$V = \int_{s_a}^\infty q(s)[V(s) - I]\, ds. \tag{3}$$

Interpretation. With no old debt, shareholders capture the full gain from future investment, so they take every positive-NPV project.

3.4 Equation (4): short-maturity risky debt

If debt matures after the state is revealed but before the investment decision, then bondholders can effectively take over in bad states and exercise the investment option whenever it is worthwhile. Debt value is

$$V_D = \int_{s_a}^{\infty} q(s)\, \min\{V(s) - I, P\}\, ds. \tag{4}$$

Interpretation.

- If $V(s) - I \geq P$, debt is paid in full.
- If $V(s) - I < P$, bondholders seize the upside up to the available asset value.
- Crucially, short-maturity debt does *not* distort the investment decision in this simple setup. The option is still exercised whenever it has positive NPV.

3.5 Equation (5): long-maturity risky debt and underinvestment

Now suppose the debt matures only *after* the investment option expires. Shareholders must contribute I today but know that some of the gain will go to old creditors. Then they invest only if

$$V(s) \geq I + P. \tag{4}$$

Let s_b satisfy $V(s_b) = I + P$. Firm value becomes

$$V = \int_{s_b}^{\infty} q(s)[V(s) - I]\, ds. \tag{5}$$

Interpretation.

- The relevant hurdle rate becomes $I + P$, not I.
- States with $I \leq V(s) < I + P$ are positive-NPV for the *firm* but not attractive to *equity*.
- That wedge is the paper's central underinvestment region.

3.6 Equation (6): debt value under long-maturity debt

If the firm does not invest, creditors get nothing because the option expires worthless. If the firm does invest, debt is paid P. Therefore

$$V_D = \int_{s_b}^{\infty} P\, q(s)\, ds. \tag{6}$$

Interpretation.

- Debt value increases with P only up to a point.

- A larger promised payment makes equity less willing to invest, shrinking the set of states in which creditors get paid.
- Hence debt value is hump-shaped in P.

3.7 Figure-3 logic: maximum debt available is not optimal debt

Equations (5) and (6) imply two distinct objects:

- the debt value-maximizing promised payment, which gives the *maximum amount lenders will advance*, and
- the firm value-maximizing promised payment, which is what shareholders should choose.

In the no-tax baseline, firm value falls monotonically with P, so the optimum is zero risky debt even though positive debt can still be priced.

Interpretation. Debt capacity is not the same as optimal leverage.

3.8 Equation (8): how debt changes the payoff to incremental investment

In the multiperiod generalization, Myers writes debt value as a function of firm value and risk,

$$V_{D,t} = f_t\Big[V_t, \sigma_t^2\Big], \tag{5}$$

where $\sigma_t^2 \equiv \sigma^2(\tilde{V}_{t+1}/V_t)$. Then the effect of incremental investment on equity value is

$$\frac{dV_{E,t}}{dI_t} = \frac{dV_t}{dI_t}\left(1 - \frac{\partial f_t}{\partial V_t}\right) - \frac{\partial f_t}{\partial \sigma_t^2}\left(\frac{\partial \sigma_t^2}{\partial I_t}\right). \tag{8}$$

Interpretation.

- The first term says that when debt is risky, some of the gain from higher firm value goes to bondholders.
- The second term captures the effect of investment on asset risk. If investment changes volatility, it changes debt value too.

3.9 Equation (9): the transfer to bondholders

Myers defines the leakage from shareholders to bondholders as

$$\begin{aligned} Z_t &\equiv \frac{dV_t}{dI_t} - \frac{dV_{E,t}}{dI_t} \\ &= \frac{dV_t}{dI_t}\frac{\partial f_t}{\partial V_t} + \frac{\partial f_t}{\partial \sigma_t^2}\frac{\partial \sigma_t^2}{\partial I_t}. \end{aligned} \tag{9}$$

Interpretation.

- Appropriate investment incentives require $Z_t = 0$.
- If debt is risky and investment does not change risk, then $\partial f_t / \partial V_t > 0$ and $Z_t > 0$.
- Shareholders then underinvest because they do not capture the full marginal benefit.

3.10 Risk-shifting appears as a secondary possibility

If incremental investment increases asset risk enough, the second term in (9) can offset or even dominate the first. Then the transfer to bondholders can shrink or turn negative.

Interpretation. Myers's framework naturally nests the idea that equity holders may prefer excessively risky projects. Underinvestment is the main message, but the model also points toward risk-shifting when new projects raise volatility.

3.11 Discrete multiperiod case: Figure 4

In the discrete version, the firm exercises an option only if

$$\Delta V(s) \geq I(s) + \Delta V_D(s), \tag{6}$$

where:

- $\Delta V(s)$ is the increase in total firm value from investment,
- $I(s)$ is the discretionary outlay,
- $\Delta V_D(s)$ is the capital gain to bondholders if the project is taken.

Interpretation. Even when the project has positive NPV for the firm, it may be rejected because too much of that NPV accrues to old debt.

3.12 Appendix equation (A.1): debt with corporate tax shields

When interest is tax deductible and the full promised interest payment can be deducted, firm value in the simple setup becomes

$$V = \int_{s_b}^{\infty} [V(s) - I] q(s)\, ds + T(P - V_D) \int_0^{\infty} q(s)\, ds, \tag{A.1}$$

where s_b is defined by

$$V(s_b) = I + P - T(P - V_D). \tag{7}$$

Interpretation. Tax shields lower the effective exercise threshold because debt now produces a tax benefit. But the basic underinvestment force remains.

3.13 Appendix equation (A.2): debt value with tax shields

Debt value becomes

$$V_D = \int_0^{s_b} T(P - V_D) q(s)\, ds + \int_{s_b}^{\infty} P q(s)\, ds. \tag{A.2}$$

Interpretation. Bondholders get the tax shield in bad states and the promised payment in good states.

3.14 Appendix equation (A.3): capped tax deductibility

If tax deductions are limited by a maximum rate R, then

$$V = \int_{s_b}^{\infty} [V(s) - I]q(s)\,ds + \min\{RTV_D,\, T(P - V_D)\} \int_0^{\infty} q(s)\,ds. \tag{A.3}$$

Interpretation.

- At low leverage, tax shields can dominate and make positive debt optimal.
- At high leverage, distorted future investment dominates.
- Therefore firm value is hump-shaped in leverage even when debt carries a genuine tax advantage.

4 Identification and credibility (what makes the model persuasive)

4.1 Why the paper is so striking

The paper is persuasive because it derives a major cost of debt in a setting where many usual frictions are absent. Myers assumes perfect and complete capital markets and strips away bankruptcy costs in the baseline. If debt still causes trouble there, the mechanism is fundamental rather than ad hoc.

4.2 What exactly creates the agency problem

Three ingredients are essential.

- The firm has valuable future opportunities that require fresh discretionary spending.
- Existing risky debt is outstanding when the future decision arrives.
- Equity holders choose whether to invest, but some of the upside accrues to debt.

Interpretation. Limited liability and the timing of claims are enough to create the wedge. The issue is not that creditors seize control too early; it is that shareholders invest too little because they must share the gain.

4.3 Why real options are the right lens

Section 3 broadens the concept of "growth opportunities." Many real assets are valuable only because the firm can continue to spend money on them later – maintenance, advertising, labor, sales effort, and technology adoption all fit this logic.

Economic lesson. Debt overhang is not confined to a narrow R&D story. It applies whenever continuation value depends on future discretionary outlays.

4.4 Why ex ante contracting is hard

Myers carefully argues that one cannot solve the problem simply by writing a contract that requires value-maximizing future investment.

- **Limited liability:** shareholders cannot be forced costlessly into future assessments.
- **Verifiability:** the NPV of future projects is rarely objective and contractible.
- **Manipulability:** management can downplay opportunities or dissipate them.

Interpretation. The whole point is that "invest whenever NPV is positive" is not an enforceable covenant.

4.5 Why renegotiation is only a partial fix

If a good project would otherwise be passed up, creditors and shareholders both gain from renegotiation. But that does not make the problem disappear.

- Renegotiation is costly.
- Creditors need information about project value.
- Monitoring to obtain that information duplicates management functions.

Interpretation. Ex post bargaining reduces the distortion but adds monitoring and negotiation costs that are themselves capital-structure costs.

4.6 Short maturity helps, but it is not free

Borrowing short can reduce the underinvestment problem because debt that matures before the investment decision can be rolled over or renegotiated. But this requires an ongoing, flexible creditor relationship.

Interpretation. Short-term debt is not magic. It works because it creates repeated opportunities to reset claims before major investment decisions occur.

4.7 Why mediation, reorganization, and bankruptcy law matter

Myers interprets bankruptcy or reorganization as a social technology for mediation and fact-finding. These institutions can help resolve conflicts between debtors and creditors when distress appears.

Interpretation. Bankruptcy is not just liquidation. It can be viewed as a mechanism to reduce costly bilateral bargaining under distress.

4.8 Dividend covenants and monitoring are rational, not arbitrary

The paper gives a clean rationale for restrictive covenants.

- Dividend restrictions can stop shareholders from walking away with cash that ought to be reinvested.
- Monitoring and covenants are costly, but shareholders optimally choose them because they lower the incidence of suboptimal investment.

Interpretation. "Restrictive" covenants are not simply lender overreach; they are devices chosen by the firm's residual claimants to reduce agency costs of debt.

4.9 Reputation as a solution

Myers explicitly notes that an honest firm can announce and follow a policy of taking positive-NPV projects even when creditors would share in the gain.

Interpretation. Reputation can soften the problem, especially for long-lived firms. But the need for reputation itself confirms the existence of the underlying temptation.

4.10 Secondary markets help only under enforceable covenants

If an asset can be sold in a secondary market, the liquidation option raises value. But without enforceable restrictions on payouts, shareholders may prefer to liquidate and run when the asset sale value is high enough.

Interpretation. Marketability of assets can either raise or lower debt capacity depending on whether creditor protections are enforceable.

4.11 Why target debt ratios and maturity matching emerge naturally

The model gives theoretical content to two common corporate-finance practices.

- **Target debt ratios:** tying borrowing to assets in place reduces the transfer to old bondholders from new investment.
- **Maturity matching:** debt repayment schedules should track the decline in the value of assets already in place.

Interpretation. These are not crude rules of thumb; they are practical responses to the debt-overhang problem.

4.12 Why diversification is ambiguous

Section 4.3 shows that combining two option-bearing assets with two debt burdens can produce opposite effects. One project may rescue another, or one may drag another down.

Interpretation. Diversification lowers variance mechanically, but it also changes which projects are worth exercising once debt claims are pooled. So debt capacity is not mechanically increasing in diversification.

4.13 Main limitations

The paper is a *partial* theory of corporate borrowing.

- It does not deliver a full dynamic model of debt policy.
- It abstracts from many other motives for debt and equity choice.
- It relies on imperfect real-asset markets and costly contract enforcement even while assuming perfect financial markets.
- It does not provide a direct empirical implementation of V_G/V.

Interpretation. These are real limitations, but the paper's purpose is narrower and still extremely powerful: isolate one deep force limiting debt.

4.14 Why the paper has lasted

The paper lasts because it converts a vague intuition – "debt may make firms too cautious" – into a precise value-based mechanism. Once written this way, it generates concrete predictions for leverage, maturity, covenants, and financing of growth firms.

5 Tables and figures

This is a **theory paper**, so there are no regression tables. The crucial evidence is in the paper's diagrams, balance-sheet examples, and one combinatorial table in Section 4.3.

5.1 Figure 1: the all-equity investment rule

Read:

- The horizontal line is the required outlay I.
- The upward-sloping schedule is $V(s)$.
- The firm invests for states $s \geq s_a$, where $V(s_a) = I$.

Economic message. Under all-equity financing the firm behaves efficiently: every state with positive net present value is accepted.

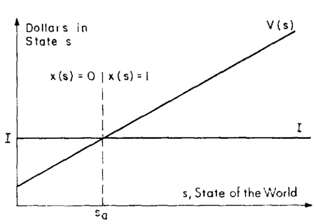

Fig. 1. The firm's investment decision under all-equity financing as a function of the state of the world, *s*, at the decision point.

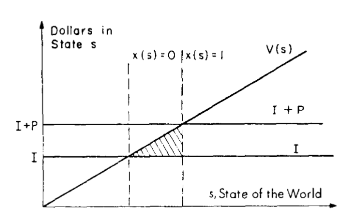

Fig. 2. The firm's investment decision with prior debt financing as a function of the state of the world, *s*, at the decision point.

5.2 Figure 2: debt pushes the cutoff from I to $I + P$

Read:

- A second horizontal line appears at $I + P$.
- Equity now invests only for $s \geq s_b$.
- The shaded triangle between I and $I + P$ is lost value.

Economic message. This is the cleanest visual statement of debt overhang. The project is good for the firm in the shaded region but bad for equity because debt captures too much of the gain.

5.3 Figure 3: firm value versus debt value as promised payment rises

Read:

- The all-equity firm value is the upper horizontal benchmark.
- Actual firm value V declines with the promised payment P.
- Debt value V_D is hump-shaped: it rises at first, reaches a maximum, then falls.

Economic message. Two ideas are packed into one picture.

- The maximum debt markets will price is finite and strictly less than firm value.
- Offering a higher promised payment can *reduce* the amount the firm can borrow.

This is Myers's credit-rationing result.

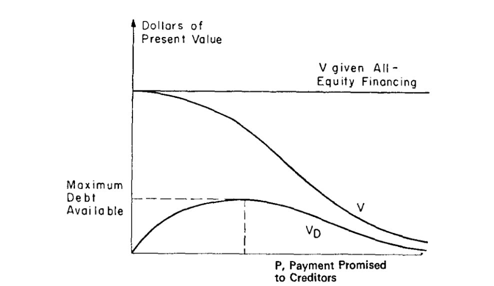

Fig. 3. Firm and debt values as a function of payment promised to creditors.

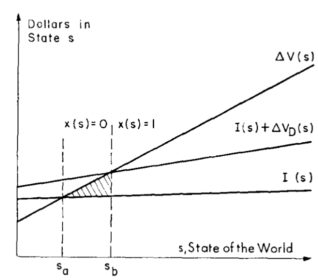

Fig. 4. The firm's investment decision with prior debt financing – multiperiod case. $I(s)$ = discretionary outlay; $\Delta V(s)$ = increase in firm value if outlay is made ($x(s) = 1$); $\Delta V_D(s)$ = increase in debt value if outlay is made, including debt service in t.

5.4 Figure 4: multiperiod underinvestment

Read:

- The project's gain to the firm is $\Delta V(s)$.
- Equity compares that gain with the cost $I(s)$ plus the bondholder capital gain $\Delta V_D(s)$.
- The shaded region again represents positive-NPV states that are rejected.

Economic message. Debt overhang is not a one-shot curiosity. It repeats in dynamic settings and carries backward and forward spillovers because current underinvestment lowers future firm value and changes future debt risk.

5.5 Table 1: diversification may help, hurt, or do nothing

Read:

- The table classifies all combinations of two option values relative to their promised payments.
- In some boxes a merger rescues a project that otherwise would not be taken.
- In other boxes one project drags the other down because the pooled debt burden becomes too high.

Economic message. There is no universal "diversification increases debt capacity" theorem in this framework.

Table 1

Investment decisions which maximize equity value when two real options and their associated debt are combined into one firm.

Net present value of option j, $V_j(s)$, relative to promised payment, P_j, on debt issued against project j	Net present value of option i, $V_i(s)$, relative to promised payment, P_i, on debt issued against project i: $V_i(s) \geqq P_i$	$P_i > V_i(s) \geqq 0$	$V_i(s) < 0$
$V_j(s) \geqq P_j$	(1) Take both $DV_i = 0$ $DV_j = 0$	(2) Take both if $V_i(s)+V_j(s)-P_i-P_j \geqq 0$. Otherwise reject both. X_i may = 1 (0 before); $DV_i \geqq 0$, X_j may = 0 (1 before); $DV_j \leqq 0$.	(3) Reject i, take j if $V_j(s)-P_i-P_j \geqq 0$. $DV_i = 0$, X_j may = 0 (1 before); $DV_j \leqq 0$.
$V_j(s) < P_j$ $V_j(s) \geqq 0$	(4) Take both if $V_i(s)+V_j(s)-P_i-P_j \geqq 0$. Otherwise reject both. X_i may = 0 (1 before); $DV_i \leqq 0$, X_j may = 1 (0 before); $DV_j \geqq 0$.	(5) Reject both. $DV_i = 0$, $DV_j = 0$.	(6) Reject both. $DV_i = 0$, $DV_j = 0$.
$V_j(s) < 0$	(7) Reject j, take i if $V_i(s)-P_i-P_j \geqq 0$. X_i may = 0 (1 before); $DV_i \leqq 0$, $DV_j = 0$.	(8) Reject both. $DV_i = 0$, $DV_j = 0$.	(9) Reject both. $DV_i = 0$, $DV_j = 0$.

Note: 'Before' refers to the decisions that would be made by firms i and j acting separately.

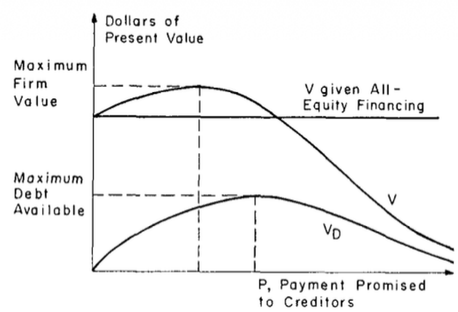

Fig. 5. Firm and debt values when debt interest is tax-deductible.

5.6 Figure 5 and the appendix: tax shields create an interior optimum

Read:

- With tax deductibility, the value of the firm first rises with debt because tax shields matter.
- Beyond a point, distorted investment dominates and firm value falls.
- The debt-value-maximizing point still lies to the right of the firm-value-maximizing point.

Economic message. Taxes justify some debt but do not rescue the "borrow as much as possible" doctrine.

5.7 Section 5 prediction set: what should be true in the data

Read:

- Assets in place should support more debt than growth opportunities.
- Capital-intensive, high-operating-leverage, and profitable assets in place should support heavier

debt financing.

- Firms should avoid borrowing against future growth opportunities.
- Sinking funds and maturity matching should be more common when protecting creditors from erosion of assets in place matters.

Economic message. The paper generates direct qualitative predictions about leverage patterns, debt contracts, and maturity structure, even without a formal empirical section.

6 Short summary

Myers develops a theory in which risky debt reduces firm value because it distorts future investment. When much of a firm's market value comes from real options or future discretionary expenditures, equity holders underinvest because part of the gain from new investment goes to existing bondholders. This makes optimal leverage lower for growth firms than for firms whose value is concentrated in assets already in place. The theory also explains why firms use target debt ratios, why they match maturities of assets and liabilities, why dividend covenants and monitoring exist, and why maximum debt capacity exceeds value-maximizing debt. With corporate taxes, the optimal debt level becomes interior, but the central underinvestment force remains.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that the amount of debt a firm should issue depends crucially on how much of firm value is an option on future action. Risky debt weakens the incentive to invest in positive-NPV opportunities whenever existing creditors capture part of the upside. Therefore firms with more growth opportunities should carry less debt than firms with more assets in place.

7.2 Economic intuition

The intuition is simple but deep. Shareholders pay the full cost of new investment, but with risky debt they do not receive the full benefit. That wedge is small when value is already sunk into assets in place. It is large when value depends on future discretionary spending. Once that is understood, low leverage by growth firms, covenant-heavy debt contracts, and maturity matching all make immediate sense.

7.3 Takeaway

This paper is foundational statements of debt overhang. It shows that even in otherwise frictionless financial markets, risky debt can make firms act too cautiously. The key question for leverage is not merely "How risky is the firm?" but "How much of firm value depends on future investment that equity might refuse to undertake once debt is in place?"